CO-1 Assignment

TITLE: Grama Jyothi Rural Electricity Co-operative Grama Jyothi is a village electricity co-operative that bills about 4,000 households every month. Ameter reader records the previous and current readings of each house. The office computes the units consumed and the bill, which has a fixed charge and an energy charge that depends on slabs of consumption (for example: the first 100 units free, the next 100 units at Rs 2.25 per unit, and every unit above 200 at Rs 4.50). The present spreadsheet has produced wrong bills: some bills are off by a few paise, a few households received a negative bill when their meter rolled over from 65535 to 0, and the tariff was typed in many places so that a tariff revision was applied only partly. The number of readings processed may grow: 1 village (4,000 houses) → 1 taluk (60,000 houses) → 1 district (800,000 houses) You are required to analyse the problem, choose suitable constants, data types and preprocessor features, implement the billing program in C and validate it, so that every bill is correct to the paisa.

Name: G R Aakaash

Reg no: 192412068

Course Code: CSA0201

Course Name: C-Programming

TASK-1

CODE:

```
#include <stdio.h>

int main()
{
   int previous, current, units;

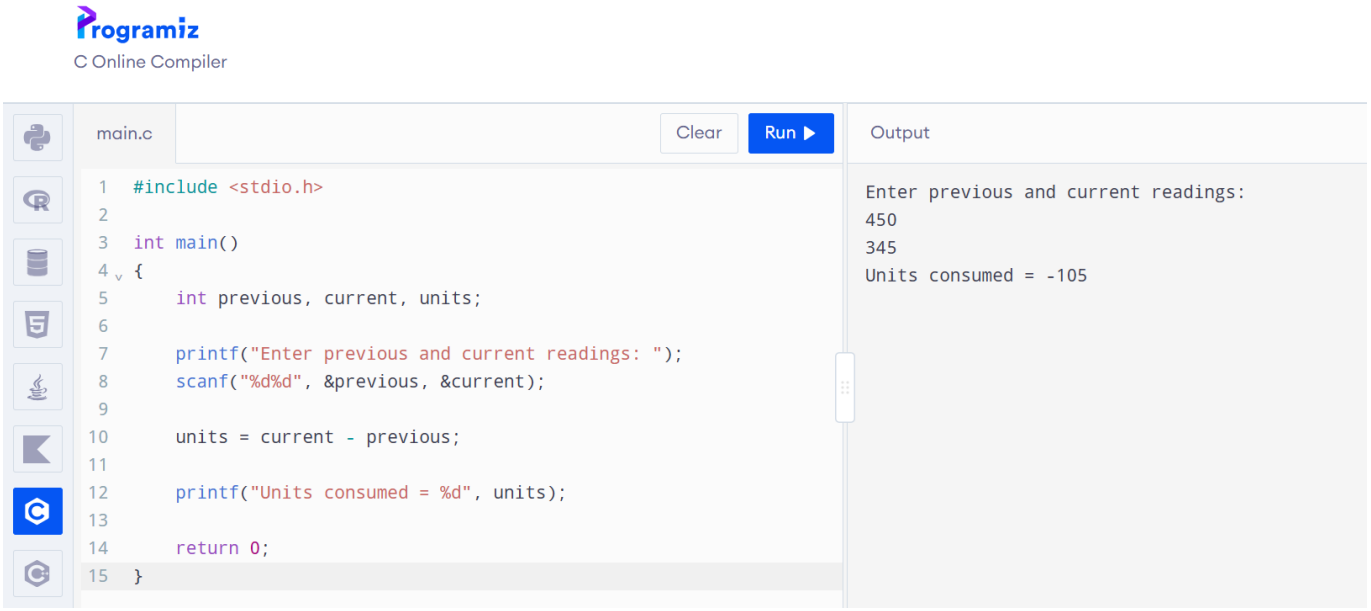
   printf("Enter previous and current readings: ");
   scanf("%d%d", &previous, &current);

   units = current - previous;

   printf("Units consumed = %d", units);

   return 0;
}
```

OUTPUT:



TASK-2

CODE:

```
#include <stdio.h>

typedef unsigned short meter_t;

unsigned int consumption(meter_t previous, meter_t current)
{
   return (unsigned short)(current - previous);
}

int main()
{
   printf("Consumption: %u units\n", consumption(1200, 1530));
   printf("Consumption: %u units\n", consumption(65400, 120));

   return 0;
}
```

OUTPUT:



TASK-3

CODE:

```
#include <stdio.h>

#define SLAB1 100
#define SLAB2 200
#define RATE1 0
#define RATE2 225
#define RATE3 450
#define FIXED 5000

long compute_bill(int units)
{
   long paise = FIXED;

   if (units > SLAB2)
      paise += (units - SLAB2) * RATE3;

   if (units > SLAB1)
   {
      int count = units < SLAB2 ? units - SLAB1 : SLAB2 - SLAB1;
      paise += count * RATE2;
   }

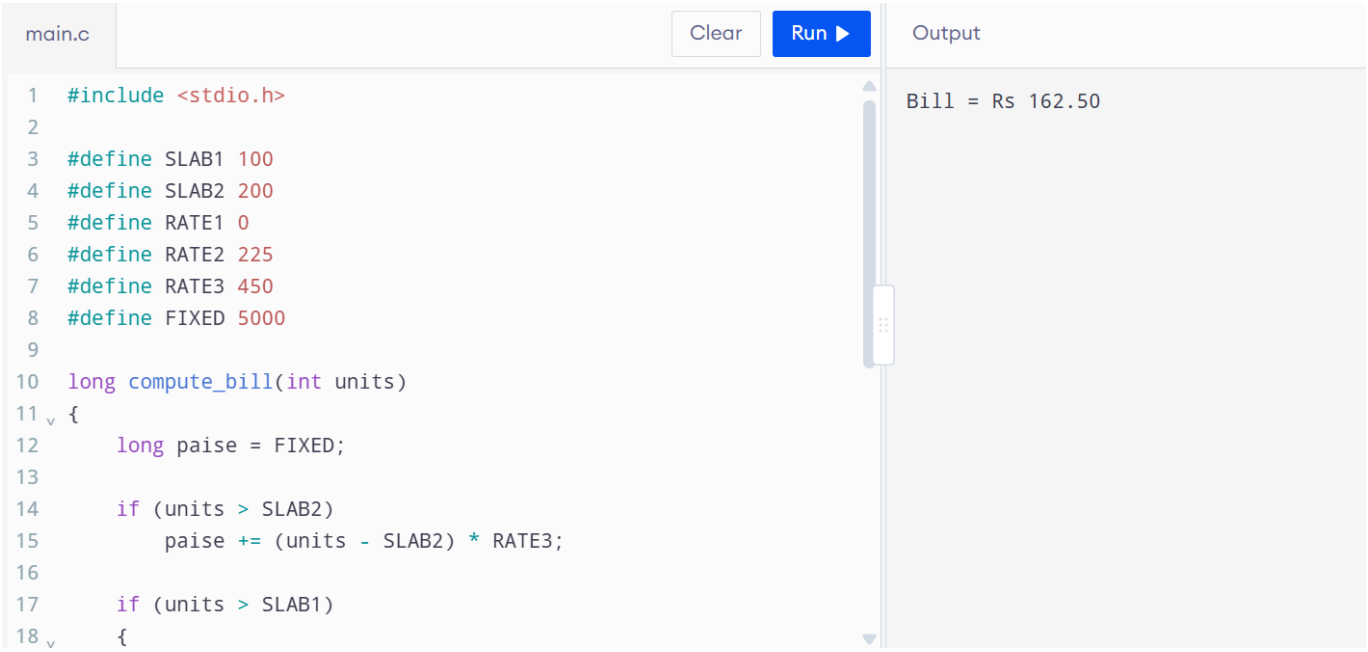
   return paise;
}
```

```
int main()
{
   int units = 150;
   long bill = compute_bill(units);

   printf("Bill = Rs %ld.%02ld", bill / 100, bill % 100);

   return 0;
}
```

OUTPUT:



TASK-4

CODE:

```
#include <stdio.h>

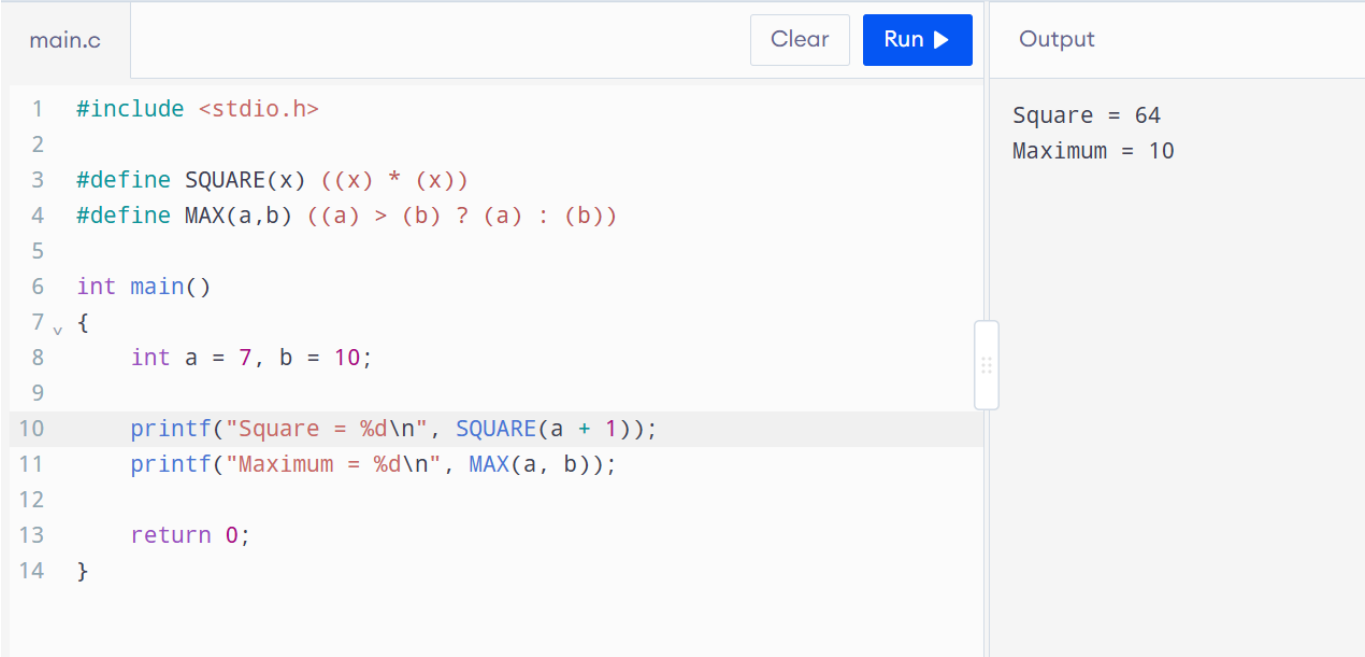
#define SQUARE(x) ((x) * (x))
#define MAX(a,b) ((a) > (b) ? (a) : (b))

int main()
{
   int a = 5, b = 10;

   printf("Square = %d\n", SQUARE(a + 1));
   printf("Maximum = %d\n", MAX(a, b));
```

```
    return 0;
}
```

OUTPUT:



TASK-5

CODE:

```
        #include <stdio.h>
#include "billing.h"

long calculate(int units)
{
    long bill = FIXED_CHARGE;

    bill += units * 225;

    return bill;
}

int main()
{
    int units = 150;

    #ifdef DEBUG
```

```
    printf("Units = %d\n", units);
    #endif

    printf("Bill = %ld paise", calculate(units));

    return 0;
}
```

OUTPUT:

```
#include <stdio.h>
#include "billing.h"

long calculate(int units)
{
    long bill = FIXED_CHARGE;

    if (units > 200)
    {
        bill += 100 * RATE2;
        bill += (units - 200) * RATE3;
    }
    else if (units > 100)
    {
        bill += (units - 100) * RATE2;
    }

    return bill;
}

int main()
{
    int units = 150;
```

```
output:
 Total Bill = Rs 162.50
```

TASK-6

CODE:

```
#include <stdio.h>

int main()
{
    int previous, current, units;

    printf("Enter readings: ");

    if (scanf("%d%d", &previous, &current) != 2)
    {
        printf("Invalid input");
        return 1;
    }

    units = current - previous;
```

```
    printf("Units = %d\n", units);

    return 0;
}
```

OUTPUT:



TASK-7

CODE: #include <stdio.h>

```
int main()
{
    int a = 5, b = 2;
    float x;

    printf("Integer division = %d\n", a / b);

    x = (float)a / b;

    printf("Float division = %.2f\n", x);

    printf("After casting = %.2f\n", (float)(a / b));

    return 0;
}
```

OUTPUT:



TASK-8

CODE:

```
#include <stdio.h>

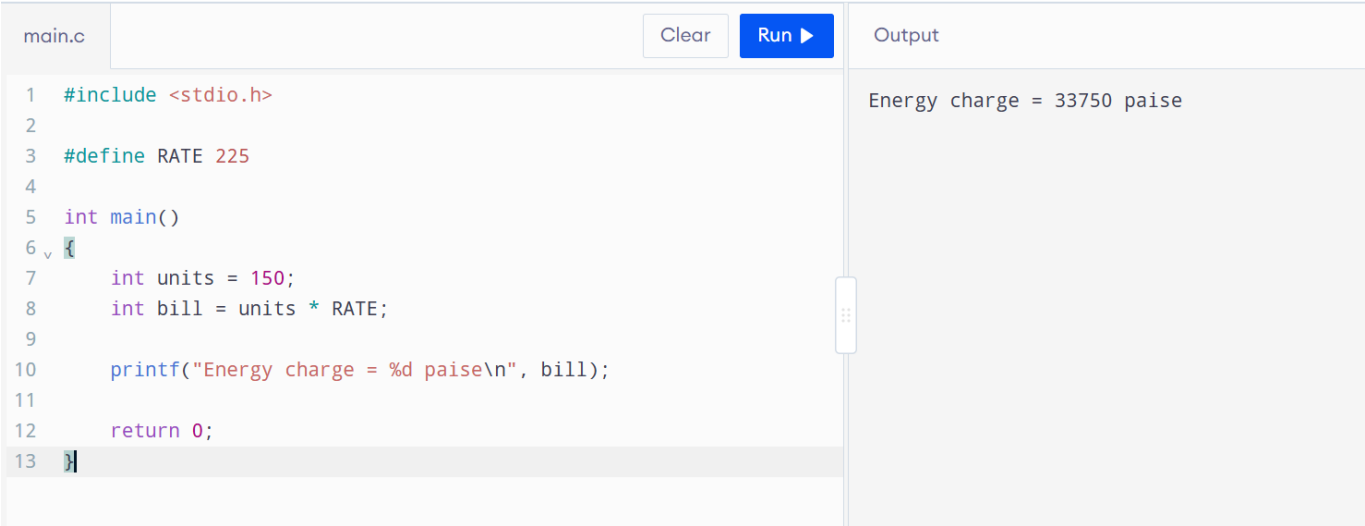
#define RATE 225

int main()
{
   int units = 150;
   int bill = units * RATE;

   printf("Energy charge = %d paise\n", bill);

   return 0;
}
```

OUTPUT:



TASK-9

CODE: #include <stdio.h>

```
#define SLAB1 100
#define SLAB2 200
#define RATE2 225
#define RATE3 450
#define FIXED 5000

int main()
{
   int previous, current, units;
   long bill = FIXED;

   printf("Enter previous and current readings: ");

   if (scanf("%d%d", &previous, &current) != 2)
      return 1;

   if (previous < 0 || current < 0 ||
      previous > 65535 || current > 65535)
   {
      printf("Invalid reading");
      return 1;
   }

   if (current < previous)
   {
      if (previous > 60000 && current < 1000)
         units = 65536 - previous + current;
```

```
        else
        {
            printf("Invalid reading");
            return 1;
        }
    }
    else
        units = current - previous;

    if (units > SLAB2)
        bill += (long)(units - SLAB2) * RATE3;

    if (units > SLAB1)
    {
        int count = units < SLAB2 ? units - SLAB1 : 100;
        bill += (long)count * RATE2;
    }

    printf("Units = %d\n", units);
    printf("Total Bill = Rs %ld.%02ld\n", bill / 100, bill % 100);

    return 0;
}
```

OUTPUT:



TASK-10

CODE: #include <stdio.h>

```
int main()
{
   float float_total = 0;
   long long paise_total = 0;

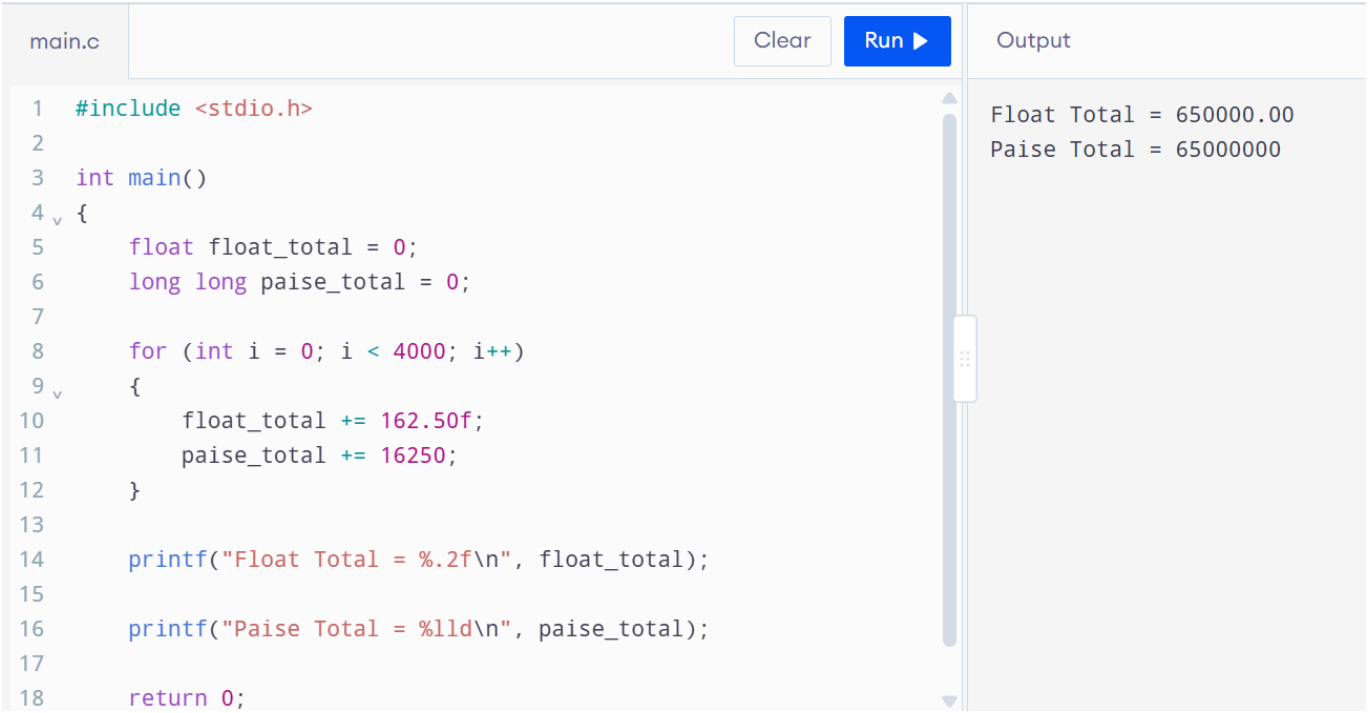
   for (int i = 0; i < 4000; i++)
   {
      float_total += 162.50f;
      paise_total += 16250;
   }

   printf("Float Total = %.2f\n", float_total);

   printf("Paise Total = %lld\n", paise_total);

   return 0;
}
```

OUTPUT:



TASK-11

CODE: #include <stdio.h>

```
#define FIXED 5000
#define RATE2 225
#define RATE3 450

long calculate_bill(int units)
{
   long bill = FIXED;

   if (units > 100)
   {
      if (units <= 200)
         bill += (units - 100) * RATE2;
      else
      {
         bill += 100 * RATE2;
         bill += (long)(units - 200) * RATE3;
      }
   }

   return bill;
}

int main()
{
   int units = 350;
   long bill = calculate_bill(units);

   printf("Units = %d\n", units);
   printf("Final Bill = Rs %ld.%02ld\n", bill / 100, bill % 100);

   return 0;
}
```

OUTPUT:



TASK-12

CODE:

```
#include <stdio.h>

int main()
{
   long long energy_wh;
   int hour, rate = 500;
   long long bill;

   printf("Enter energy in watt-hours: ");
   scanf("%lld", &energy_wh);

   printf("Enter hour (0-23): ");
   scanf("%d", &hour);

   if (energy_wh < 0 || hour < 0 || hour > 23)
   {
      printf("Invalid input");
      return 1;
   }

   if (hour >= 18 && hour < 22)
      rate = 600;  // 20% extra

   bill = (energy_wh * rate + 500) / 1000;
```

```
    printf("Electricity charge = Rs %lld.%02lld\n",
           bill / 100, bill % 100);

    return 0;
}
```

OUTPUT:

